

Partial Differential Equations

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1 Introduction

As this series of lectures and notes covers both partial derivatives and ordinary differential equations (ODEs) we will not be doing as much of a recall of those subject and instead make note that this is a generalization of the two subjects. What that looks like is a ordinary differential equation is an equation of one input and one output involving a derivative loosely speaking. And a partial derivative is loosely speaking the derivative of a multivariate function with respect to a signal variable.

Just as many ODEs are not solvable analytically, many partial differential equations (PDEs) are also not solvable analytically. Still, there are a few methods that allow us to obtain a non numerical solution. The primary go to is Fourier analysis, but this will be saved for the lectures and notes covering that topic. Instead here we will look briefly at the method of characteristic curves and separation of variables. Both of these techniques for finding solutions share the basic idea of taking an PDE and turning it into an ODE. This is because we know more about ODEs in comparison to PDEs. This is in part due to the large flexibility allowed over a multi-variable equation.

Moving to PDEs more conceptual side, similar to how ODEs are used to model a changing system PDEs are a generalization of this, being suited quite nicely to model a changing system of multiple variables. This can take form in a number of different ways, for example while an ODE can model the change in volume or concentration of a liquid over time, a PDE can model reach further into the change in the position of a wave in time. More specifically, an ODE limits us to the case where our concentration of volume is dependent on time. The PDE instead allows for position to be independent in addition to time being an independent variable.

2 Definitions

First recall the definitions of an Ordinary differential equation and partial derivatives.

Ordinary Differential Equation

Define an Ordinary Differential Equation: Let V be the vector space with basis $\{x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}\}$ and $F : V \rightarrow V$ and $G : V \rightarrow V$. Then F, G are functions of x, y and the derivatives of y . In explicit form we have;

$$F(x, y, \frac{dy}{dx}, \dots, \frac{d^{n-1}y}{dx^{n-1}}) = \frac{d^n y}{dx^n}$$

And in implicit form;

$$G(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}) = 0$$

Definition of the Partial Derivative

Let $A \subseteq \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the partial derivative with respect to the i^{th} coordinate

$$f_{x_i} = \frac{\partial f}{\partial x_i} := \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

With these definitions in mind, let's attempt to generalize our understanding of differential equations into higher dimensions!

Partial Differential Equation 1

A partial differential equation is an equation involving an unknown function of two or more variables and its partial derivatives.

Let's now rewrite this definition with an equation.

Partial Differential Equation 2

Let $A \subseteq \mathbb{R}^n$ and $u : A \rightarrow \mathbb{R}$. Suppose $D^k u$ denotes the k^{th} partial derivative of u . Then a partial differential equation in implicit form is of the form

$$F(u, x_0, x_1, \dots, x_n, D^1 u, \dots, D^k u) = 0$$

where $x_i \in A \forall 0 \leq i \leq n$

Observe F is a function from the vector space with the basis of $\{u, x_0, x_1, \dots, x_n, D^1 u, \dots, D^k u\}$ to $\mathbb{R}$.

3 Examples

We are going to apply the method of separation of variables as an introduction into PDEs, but know typically we will apply Fourier series to find solutions to PDEs in the field of engineering. For more on the use of Fourier series refer to the Fourier lecture and notes for an introduction to that topic and the Fourier and PDE lecture and notes for the marriage of the two topics.

Lets consider a classic example, the one dimensional wave equation.

Let L be the length of a uniform string and t be time. The wave deformation equation is then

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, 0 < x < L, t > 0$$

Notice this is a second order homogeneous partial differential equation of two variables. Which means we can solve it! And easily (relatively speaking)!

Applying separation of variables for PDEs we will let $u(x, t) = T(t)X(x)$ where T is a function of only t and X is a function of only x . Here it is important to note that this is a guess, and if the PDE does not have a solution of this form, when applying the solution we will not be able to cleanly separate the variables.

Lets attempt to separate the variables now. So lets find the second derivatives of our guess of u .

$$\begin{aligned} u(x, t) &= T(t)X(x) \\ \frac{\partial u}{\partial x} &= T(t)X'(x) \\ \frac{\partial^2 u}{\partial x^2} &= T(t)X''(x) \\ \frac{\partial u}{\partial t} &= T'(t)X(x) \\ \frac{\partial^2 u}{\partial t^2} &= T''(t)X(x) \end{aligned}$$

As if we recall from our understating of partial derivatives we treat other variables as a constant when taking a partial.

Plugging into the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$T''(t)X(x) = c^2 T(t)X''(x)$$

And as neither $X(x)$ nor $T(t)$ is zero as we would then not have a partial differential equation

$$T''(t)X(x) = c^2 T(t)X''(x)$$

$$\frac{T''(t)}{T(t)c^2} = \frac{X''(x)}{X(x)}$$

So our differential has two sides. One that is only of x and one that is only of t thus they must be equal to the same constant as both x, t are variables so we can write

$$\frac{T''(t)}{T(t)c^2} = k = \frac{X''(x)}{X(x)}$$

Breaking this apart into two ODEs we have

$$\frac{T''(t)}{T(t)c^2} = k$$

$$T''(t) - kc^2 T(t) = 0$$

and

$$\frac{X''(x)}{X(x)} = k$$

$$X''(x) - kX(x) = 0$$

I leave it to you as an exercise to check my findings of the solutions to both these ODEs but you can apply the method of variation of parameters to find

(hint let $k = \mu^2$) and check positive and check $+\mu^2$ and $-\mu^2$)

$$X(x) = c_1 \cos(\mu x) + c_2 \sin(\mu x)$$

and as $X(0) = X(L) = 0$ we can write the full solution for X

We know $c_1 = 0$ as the $\cos(x)$ is nonzero at $x = 0$ and applying trig (which I want you to practice by finding all the multiples to make $\sin 0$ under its period.) And let $c_2 = 1$ for simplicity

$$X(x_n) = \sin\left(\frac{n\pi}{L}x\right)$$

Now from your solution to

$$T''(t) + \left(c\frac{n\pi}{L}\right)^2 T(t) = 0$$

You will have found from your solving of this ODE

$$T_n = b_n \cos\left(c\frac{n\pi}{L}t\right) + b'_n \sin\left(c\frac{n\pi}{L}t\right)$$

which gives

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} u_n(x, t) \\ &= \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left(b_n \cos\left(c\frac{n\pi}{L}t\right) + b'_n \sin\left(c\frac{n\pi}{L}t\right)\right) \end{aligned}$$

And as you can see had we let c_2 be some other number or solved for a number we would have still had the same variability due to our other constants of integration from T , being b, b' .

In sum this is an example to give you an idea of what solving a PDE looks like analytically, but more important than being able to understand all of this is being able to recognize it. As we as engineers apply numerical methods and have computers solve our PDEs, but we need to know when we are dealing with one so that we might program our computers correctly and apply the correct numerical methods of approximation.